

Mathematical Foundations of Image Synthesis

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1 Introduction

Image synthesis is the process of generating images from mathematical and algorithmic descriptions. Modern graphics systems rely on **precise mathematical models** and efficient computation. *Even small numerical errors* can result in visible artifacts in rendered images.

Typical formulations involve scalar values such as x , indexed parameters like I_{ij} , and exponential terms e^{x^2} . These elements are combined to simulate realistic lighting behavior.

2 Core Components

2.1 Rendering Pipeline

The rendering pipeline consists of the following stages:

- Scene Definition
- Camera Configuration
- Light Placement

Important Shading Model

- Diffuse Shading
- Specular Highlights

1. Input Processing
2. Ray Construction
 - (a) How are you
 - (b) Secondary Rays
3. Intersection Evaluation
 - Object Hit Test

3 Mathematical Modeling

Light interaction with surfaces can be expressed using the following equations.

$$L = I_{ij} \cdot \cos(\theta) + R^2 \tag{1}$$

$$k = \frac{ab}{cd}$$

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x + 1)^2 \end{aligned} \tag{2}$$